

Resistance

- Depend Internally, $R = \rho \frac{L}{A}$ $\rho \rightarrow [\Omega m]$ $R \rightarrow [\Omega]$

* Conductance is opposite of Resistance.

$$G = \frac{1}{R} = \frac{A}{\rho L} = \frac{\sigma A}{L} ; \sigma = \frac{1}{\rho} \quad \sigma \rightarrow [S m^{-1}]$$

Siemens per meter

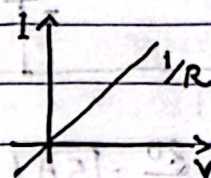
$$G \rightarrow [S] \text{ or } [U]$$

↑ mhos mho

Circuit Law

⇒ Ohm's Law: $I \propto V$; $\frac{V}{I} = \text{const} \rightarrow R$

$$\therefore I = \frac{V}{R}$$



$$* P = Vi = i^2 R = \frac{V^2}{R}$$

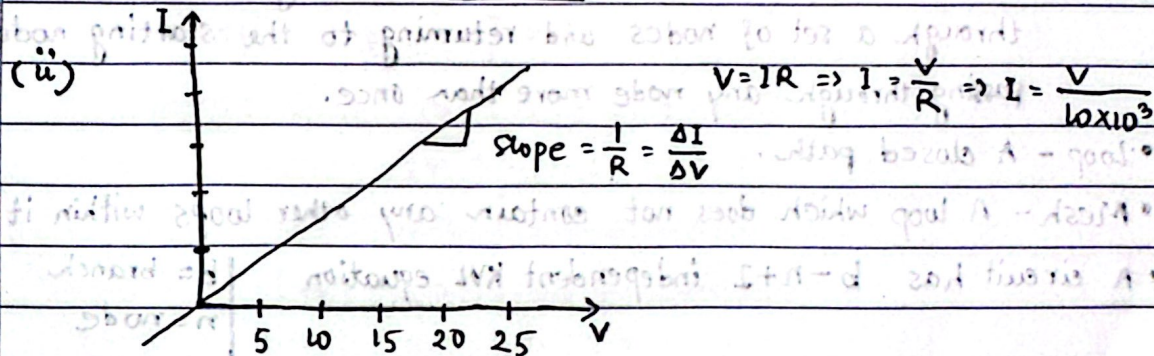
Ex:1: $i = \frac{V}{R} = \frac{30}{5000} = \boxed{0.006 A}$; $G = \frac{1}{R} = \frac{1}{5000} = \boxed{0.0002 S}$

$$P = Vi = 30 \times 5000 \times 0.006 = \boxed{0.18 W}$$

Problem 1:

(i) $V = IR = 3 \times 10^{-3} \times 10 \times 10^3 = \boxed{30 V}$; $G = \frac{1}{R} = \frac{1}{10 \times 10^3} = \boxed{1 \times 10^{-4} S}$

$$P = Vi = 30 \times 3 \times 10^{-3} = \boxed{0.09 W}$$



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Problem 2: $P = vi = i^2 R = \frac{V^2}{R}$

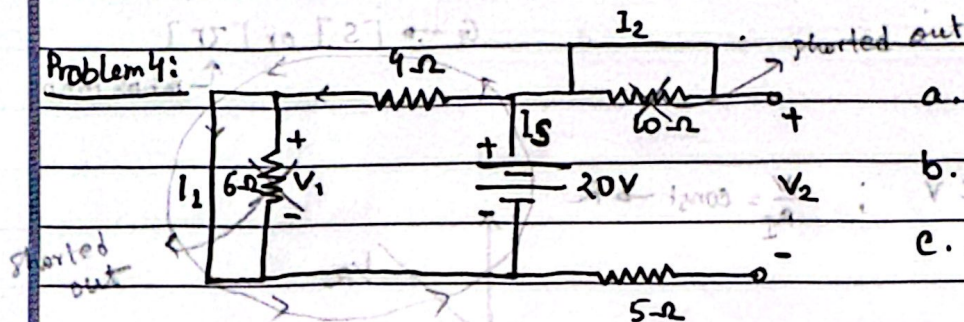
$$P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} = \frac{120^2}{60} = \boxed{240 \Omega}$$

Problem 3:

(a) $i = \frac{V}{R} = \frac{40}{100} = \boxed{0.4 A}$

(b) $i = \frac{40}{250} = \boxed{0.16 A}$

Problem 4:



a. I_1 & $I_2 = ?$

b. V_1 & $V_2 = ?$

c. $I_s = ?$

$\Rightarrow$ (a) $I_2 = \boxed{0 A}$, $I_1 = \frac{20}{4} = \boxed{5 A}$

(b) $V_2 = \boxed{0 V}$, $V_1 = |V_a - V_b| = 20V$

(c) $I_s = I_1 = \boxed{5 A}$

⇒ Circuit Terminology:

- Branch - A single path, contain one simple element, connects one node to another. Ex: Resistor, Battery
- Node - The point connecting two or more branches. Equipotential region.
- Path - Is a set of elements formed by starting at a node passing through a set of nodes. and returning to the starting node without passing through any node more than once.
- Loop - A closed path.
- Mesh - A loop which does not contain any other loops within it.

* A circuit has $b - n + 1$ independent KVL equation

$\left| \begin{array}{l} b = \text{branch} \\ n = \text{node} \end{array} \right.$

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* LOOP VS MESH

loop1: 10V, 5Ω, 2Ω (Mesh) (Independent)

loop2: 10V, 5Ω, 3Ω (Independent)

loop3: 10V, 5Ω, 2A (Independent)

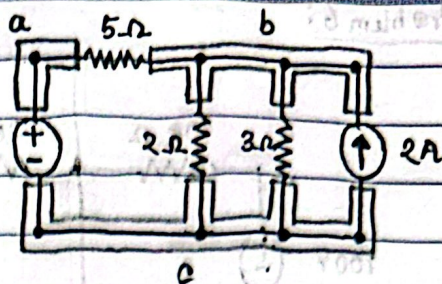
loop4: 2Ω, 3Ω (Mesh)

loop5: 2Ω, 2A

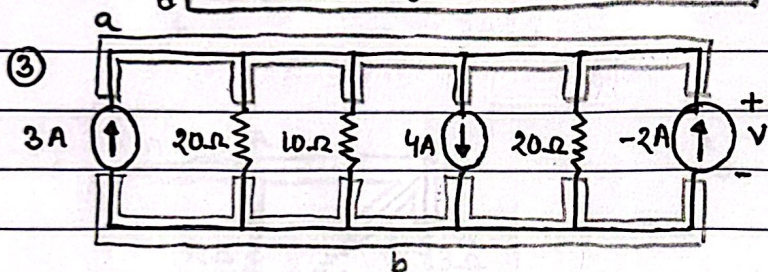
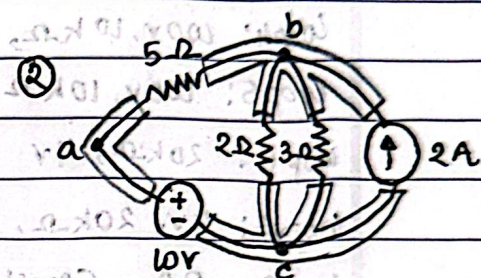
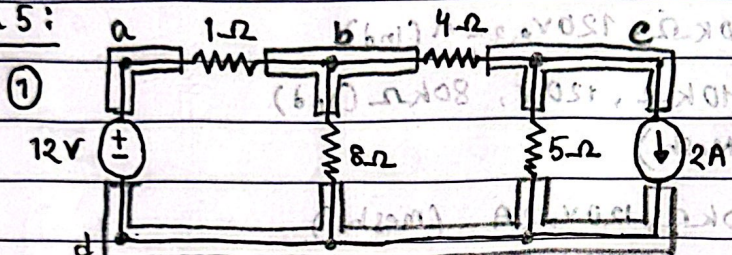
loop6: 3Ω, 2A (Mesh)

Number of meshes = Number of Independent KVL equation

* Total Number of loops and meshes in a circuit are always constant.



Problem 5:



$\Rightarrow$ (1) $\Rightarrow$ branches: $\{12V, 1\Omega, 4\Omega, 5\Omega, 8\Omega, 2A\} = 6$

nodes: $\{a, b, c, d\} = 4$

loops: loop1: 12V, 1Ω, 8Ω (Mesh)

loop2: 12V, 1Ω, 4Ω, 5Ω

loop3: 12V, 1Ω, 4Ω, 2A

loop4: 8Ω, 4Ω, 5Ω (Mesh)

loop5: 8Ω, 4Ω, 2A

loop6: 5Ω, 2A (Mesh)

loop = 6

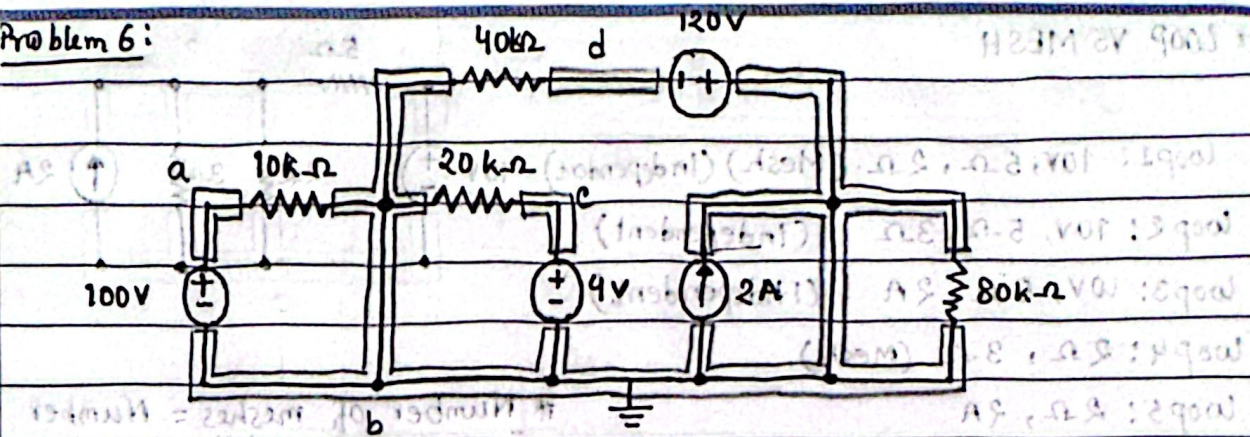
Mesh = 3

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Problem 6:



$$l = (b - n + 1) = 5$$

loop 1: 100V, 10kΩ (Ind) (Mesh)

loop 2: 100V, 10kΩ, 20kΩ, 4V (Ind)

loop 3: 100V, 10kΩ, 40kΩ, 120V, 2A (Ind)

loop 4: 100V, 10kΩ, 40kΩ, 120V, 2A (Ind)

loop 5: 100V, 10kΩ, 40kΩ, 120V, 80kΩ (Ind)

loop 6: 20kΩ, 4V (Mesh)

loop 7: 4V, 20kΩ, 40kΩ, 120V, 2A (Mesh)

loop 8: 2A (Mesh)

loop 9: 80kΩ (Mesh)

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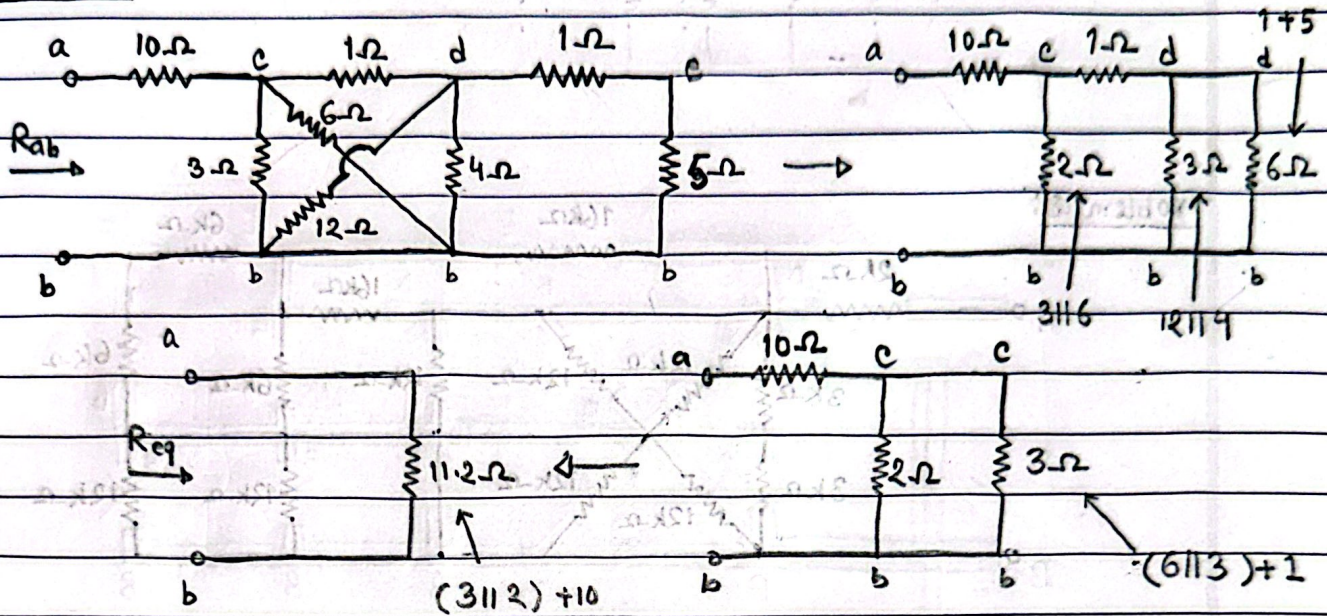
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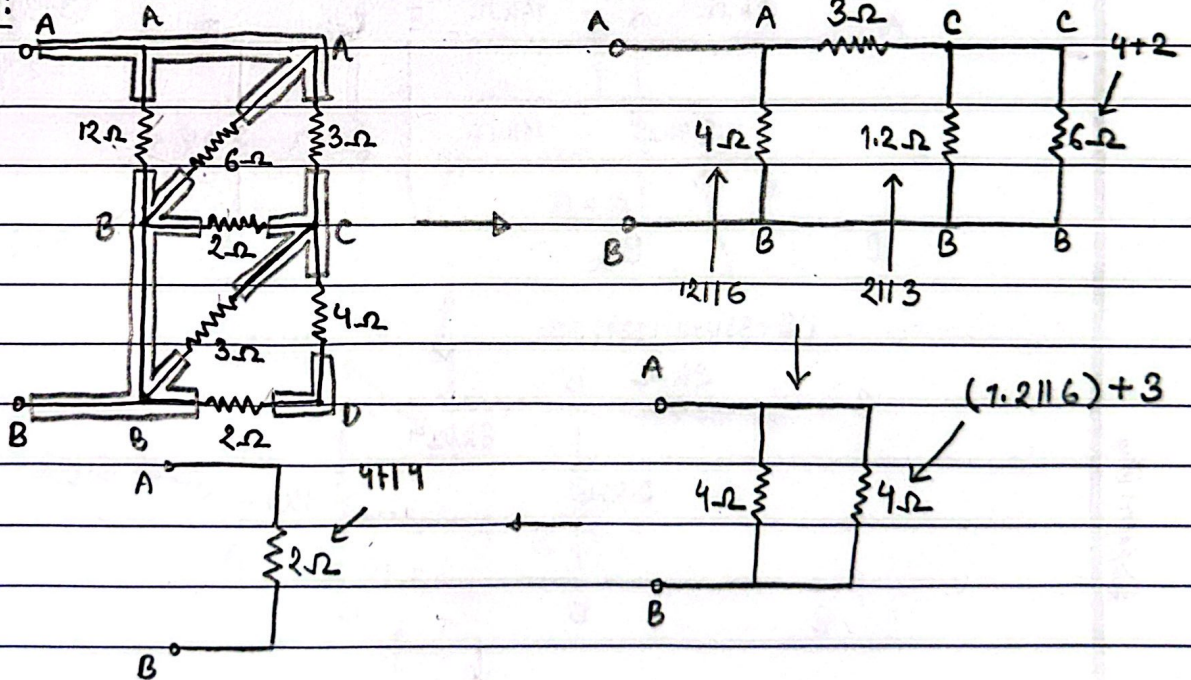
⇒ Circuit Configuration:

- Series
- Parallel
- 'Y' or 'Δ' → Wye/Delta Configuration
or

Example 2: Find R_{ab}



Problem 12:

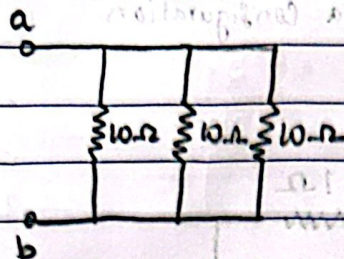
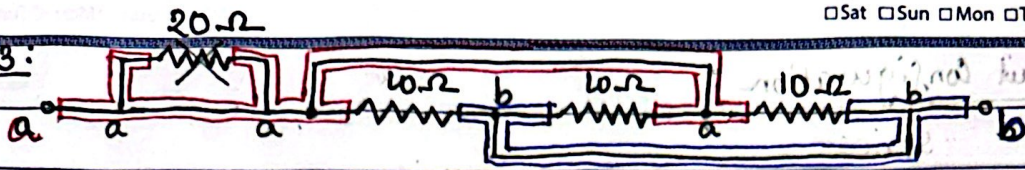


$\therefore R_{AB} = 2\Omega$

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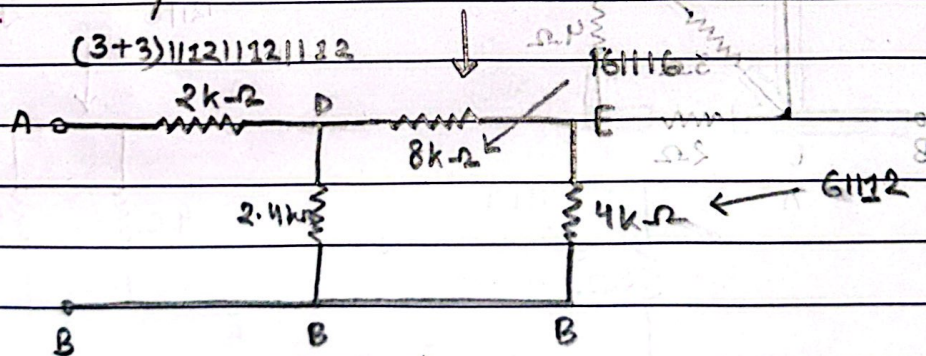
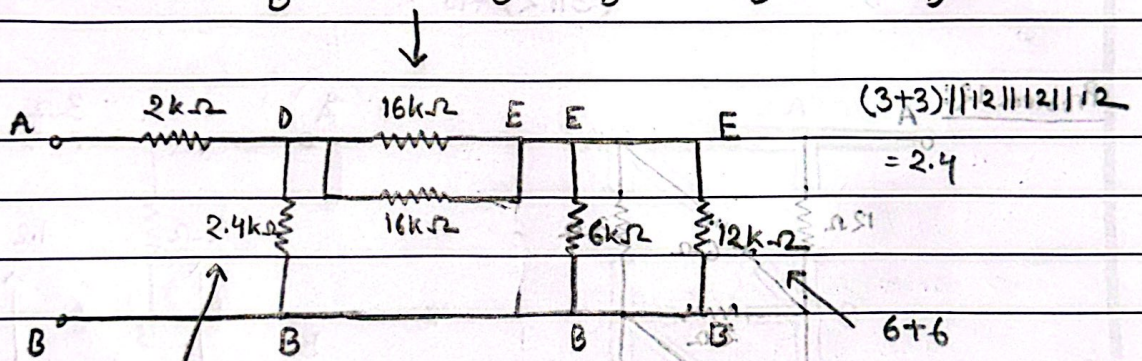
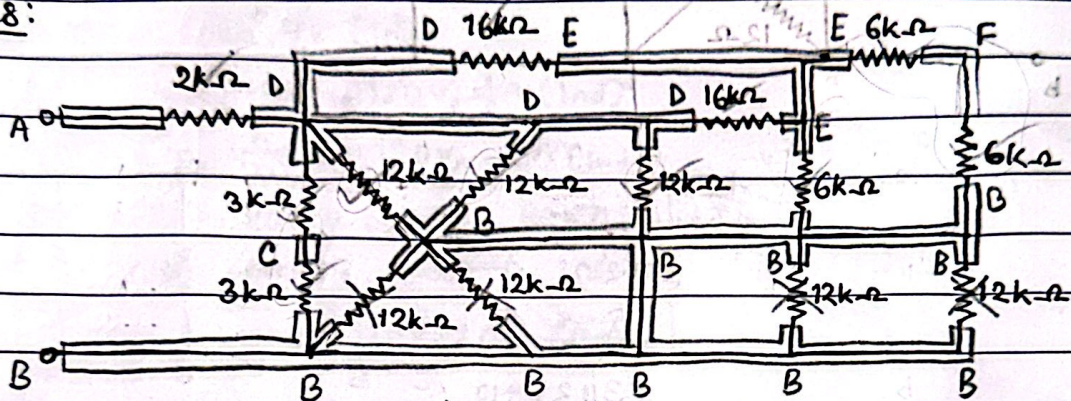
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Example 3:



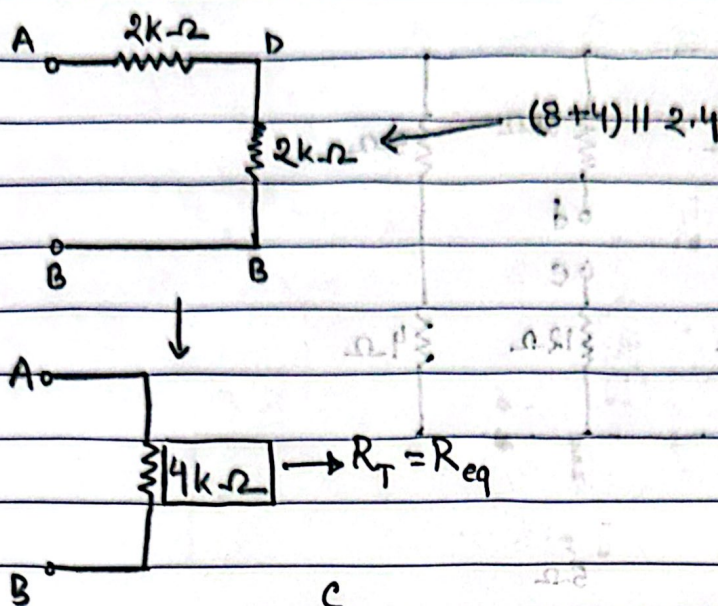
$$R_{ab} = 10 \parallel 10 \parallel 10 = \frac{10}{3} \Omega$$

Problem 28:

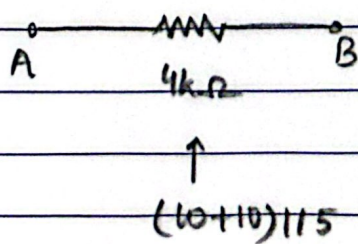
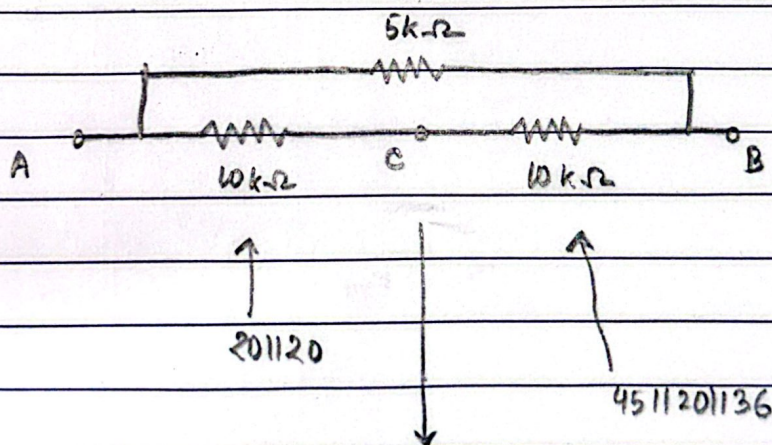
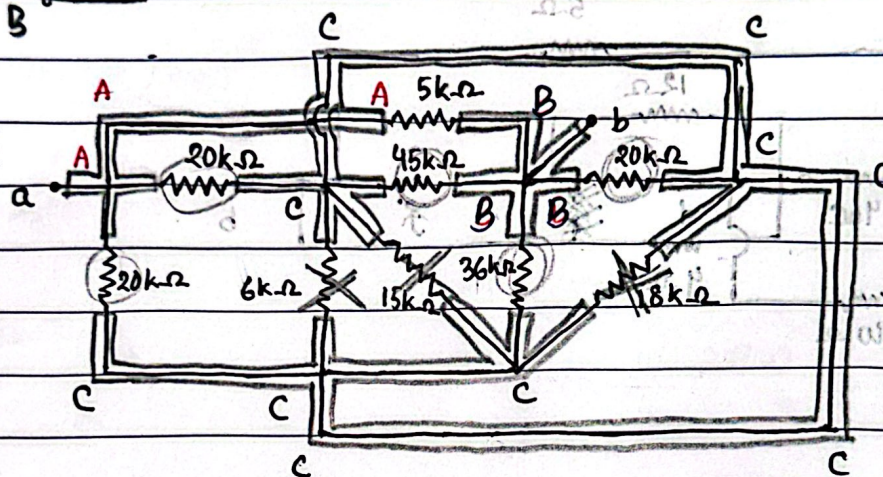


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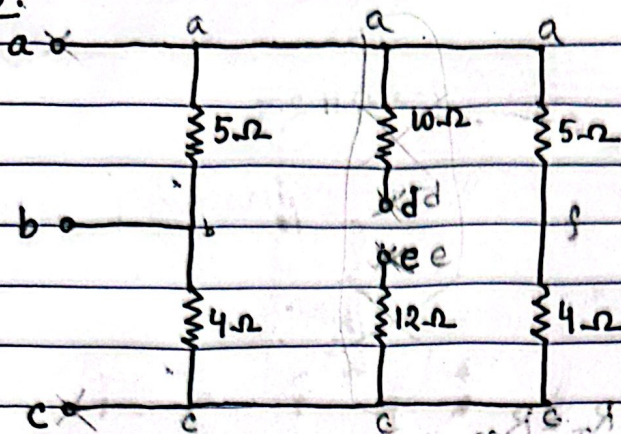
Problem 19:



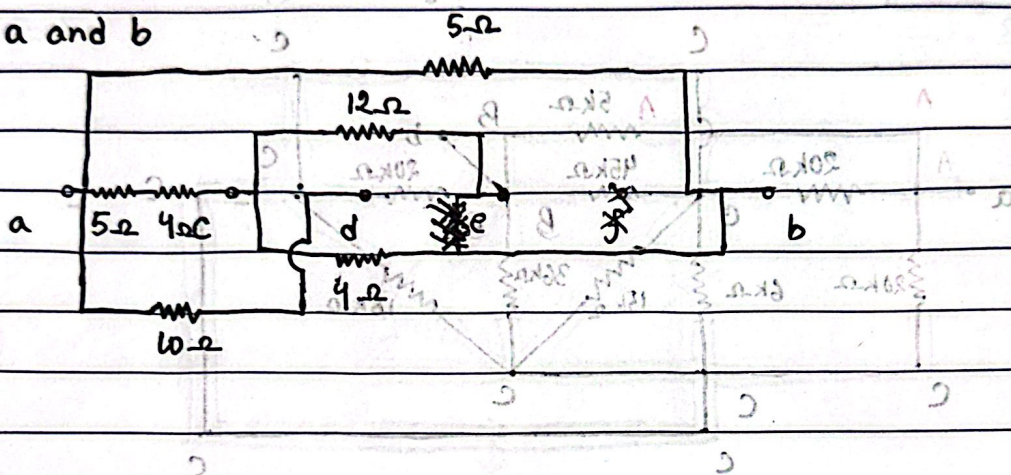
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Problem 2D:



I. a and b



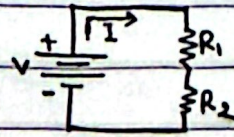
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⇒ Voltage Division Rule:

$$I = \frac{V}{R_1 + R_2}$$



$$V_1 = IR_1 ; V_2 = IR_2$$

$$\therefore V_1 = \frac{R_1}{R_1 + R_2} V ; V_2 = \frac{R_2}{R_1 + R_2} V$$

∴ If Resistors are in series,

$$V_x = \frac{R_x}{R_1 + R_2 + \dots + R_N} V$$

⇒ Current Division Rule:

IF Resistors are in Parallel

$$I_x = \frac{R_{eq}}{R_x} \times I ; I_x = \frac{(R_x)^{-1}}{(R_1)^{-1} + (R_2)^{-1} + \dots + (R_N)^{-1}} \times I$$